feel already comfortable with reading pseudo-code, you can safely skip this appendix, or just refer to it when you feel you'd need clarifications on notation.

A.1 Variables and Basics

Like for every other programming language, the first fundamental step is being able to save values and recover them: as you all know, while low level languages like `Assembly` uses *registers* as locations where you can park a value, higher level languages introduced the concept of *variables*.

Variables are just named placeholders that can be created and to which values can be assigned and later read.

Some languages, the strongly typed ones, requires each variable to accept only a certain type of values (for example integers or strings) for the whole life of a variable. Other languages (loosely typed) will allow variables to hold any value without restricting them to a single type. The loosely-typed approach also removes the need to declare variables before assigning them: a variable will be automatically created when it is first assigned (but note that using its value before assigning it will result in an error, in all loosely-typed languages).

For pseudo-code, using a loosely-typed approach is pretty much natural: as mentioned, we would like to abstract as much as possible from implementation details.